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## Task-03 : (The Overfitting Astronaut)

### Step1:

My Model Rule; If →

I will classify a photo as a follow sign (YES) only if all the following conditions are satisfied:

1) The astronaut is wearing a blue shirt.
2) The sign is white.
3) The word FOLLOW is written in black.
4) The word FOLLOW is upright.

Then:

YES → Follow sign.

ELSE:

NO → Not a Follow sign.

### Step:2 (Testing my rules on new photos)

| Test Photo | My prediction | Actual result |
|---|---|---|
| A | NO | Follow |
| B | NO | Follow |
| C | NO | NOT Follow |
| D | NO | Follow |

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Explanation:

⇒ A: NO, because the astronaut wears a white EVA suit and the world is written in red.

⇒ B: NO, because the word FOLLOW is upside down.

⇒ C: NO, because the sign says STOP instead of FOLLOW.

⇒ D: NO, because the astronaut wears a white EVA suit and holds an arrow instead of the specified sign.

Step: 3 (Reflection Questions' Answers)

1) My rule correctly classified only 1 out of 4 test photos. It correctly identified photo C as NOT follow but failed on photos A, B and D.

2) My rule focused too much on the astronaut's blue shirt, white sign and black writing instead

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of the actual meaning of the sign. This created data bias because all six training photos showed astronauts wearing blue shirts. As a result, my model learned irrelevant features instead of the actual concept of a Follow sign.

3) My rule required the word FOLLOW to be upright and readable. However, photo B had the same word upside down. This is an example of overfitting because my model memorized the specific appearance and orientation of the training examples instead of learning the general concept.

4) I would collect three more diverse photos:

i) one astronaut wearing a white EVA suit holding a FOLLOW sign written in red.

ii) one astronaut holding an upside-down or tilted FOLLOW sign.

iii) one astronaut holding a hand-drawn arrow pointing forward.

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These examples would help the model recognize different appearance, orientations and sign designs.

5) The model might fail to recognize the follow sign because it learned to associate blue shirts with the correct result. This would reduce its accuracy during competition. To improve it, I would create a diverse and balanced training dataset containing astronauts in different suits, different sign colors, different orientations, lighting conditions, and backgrounds. This would help the model generalize better to unfamiliar situations.

conclusion: This task helped me understand supervised learning, overfitting, data bias and generalization. I learned that a good machine learning model should focus on important features instead of memorising unnecessary details. A diverse training dataset helps a model perform better on new images.